Individual report

BSC HONS ETHICAL HACKING AND CYBER SECURITY SOFTWARICA COLLEGE OF IT AND E- COMMERECE

ST4059CEM LEGAL AND ETHICAL FOUNDATIOONS IN CYBER SECURITY

Student ID Number: 17108234 **Coventry id: 250388**

Submitted by: **submitted to:**

Swarnim Limbu **MR. Ganesh Bhusal**

ABSTRACT

In this assignment we will learn various cyber law such as GDPR, ETA 2063 and many more etc there are various thing that we might learn in this lesson because there are 4 different task that each of them covers differently such as Task 1 where task 1 covers ethical and legal frame work the Task 2 covers the ethical dilemmas where we talk about the benefit & disadvantage of AI covering it with Utilitarianism and Deontological Ethics the task 3 covers the importance of leadership & its role in organization and task 4 covers the ethical portfolio and Self-Reflection on Cybersecurity Skills and Ethical Growth.

Table of Contents

TASK 1: Comparative Cybersecurity Legal Frameworks 4

1.1 Comparative Legal Ecosystems 4
1.2 Framework Comparison: Cybercrime, Data Privacy and Civil Rights 5
1.3 Case Study – Nepal: Mangaluru Fraud 6
1.4 Case Study – UK: WannaCry Ransomware 7
1.5 Legal Limitations and Strategic Reform 8
1.6 Proposed Reform Strategies.....8

TASK 2: AI Threat Detection – Ethical Dilemmas 10

2.1 Introduction..... 10
2.2 Technology in Focus 10
2.3 Ethical Dilemma 1: Algorithmic Bias and Discrimination 10
2.4 Ethical Dilemma 2: Surveillance vs Privacy 11
2.5 Ethical Evaluation: Utilitarianism 11
2.6 Ethical Evaluation: Deontological Ethics 12
2.7 Conclusion 19

TASK 3: Leadership, Security Culture and Governance.....13

3.1 Leadership and Organizational Security Culture..... 13
3.2 Weak Leadership Case Study – Equifax 2017 Breach 13
3.3 Strong Leadership Case Study – Microsoft and IBM 13
3.4 Impact of Leadership on Security Culture 14
3.5 Integrated Cybersecurity Ethics and Compliance Policy..... 14
3.6 Proposed Cybersecurity Ethics and Governance Framework 15
3.7 Conclusion 19

TASK 4: Self-Reflection and Professional Ethical Portfolio 16

4.1 Self-Reflection on Cybersecurity Skills and Ethical Growth 16
4.2 Digital Forensics 16
4.3 Legal and Regulatory Compliance 17
4.4 Ethical Incident Response and Data Privacy 17
4.5 Building a Professional Ethical Portfolio 18
4.6 Ethical Decision-Making Under Pressure 18
4.7 Legal and Regulatory Compliance Documentation 18
4.8 Technical Depth and Demonstration 18
4.9 Ethical Branding and Reputation Management 19
4.10 Conclusion 19

Table of Figures

TASK 1 – Comparative Cybersecurity Legal Frameworks

Figure 1: Comparative Overview of Nepal and UK Cybersecurity Legal Frameworks 4
Figure 2: Nepal Cybercrime Case – Mangaluru Fraud Network Attack Flow 6
Figure 3: WannaCry 2017 Ransomware Attack – UK Legal Response Analysis 7
Figure 4: NIST Cybersecurity Framework – Proposed Reform Model for Nepal 8

TASK 3 – Leadership, Security Culture and Governance

Figure 1: The Role of Leadership in Building Organizational Security Culture 13
Figure 2: Task 3 Assessment Criteria – Leadership, Security Culture & Policy Development..... 14
Figure 3: Proposed Unified Cybersecurity Ethics & Governance Framework 15

TASK 4 – Self-Reflection and Professional Ethical Portfolio

Figure 1: Digital Forensics – Evidence Collection and Cyber Investigation 17
Figure 2: Legal Compliance Frameworks – GDPR and Nepal's Electronic Transactions Act 206318
Figure 3: Building a Professional Ethical Portfolio in Cybersecurity 19

TASK 1:Comparative Cybersecurity Legal Frameworks and Global Jurisprudence

Comparative Legal Ecosystems:

Comparative Cybersecurity Legal Frameworks: Nepal vs UK

	NEPAL (ETA 2063 / IT Bill 2075)	UK (CMA 1990 / DPA 2018 / NIS)
Cybercrime Law	ETA 2063 - Criminal offences	Computer Misuse Act 1990
Data Protection	Limited - No dedicated authority	GDPR + Data Protection Act 2018
Critical Infrastructure	Weak / Underdeveloped	NIS Regulations 2018
Jurisdiction	Territorial limitations	Extraterritorial jurisdiction
Enforcement Body	Nepal Police / NTA	ICO + NCSC + GCHQ
Regulatory Depth	Punitive-focused	Preventive + Regulatory

***Figure 1**: Comparative Overview of Nepal and UK Cybersecurity Legal Frameworks*

Cyber governance reflects nation's balance between security, economic development and civil liberties. Nepal cyber called ETA 2063 was created to make digital documents and signatures a form of security to identify originality but this ETA 2063 lacks soe confidentiality such as it only focuses on punishing people after a crime happens rather than preventing attacks in the first place by making organizations follow security rules to protect data before anything goes wrong. So, to replace this law IT BILLING 2075 was introduced to modernized Nepal cyber framework by expanding the law of cybercrime, strengthening investigative powers and introducing regulatory oversight mechanisms.

In contrast, the UK has built a multiple protection layer of systems along with mature cybersecurity eco-system. The computer Misuse Act 1990 (CMA) punishes people who hack into computers, alter information without permission, or damage or shut down computer systems. The CMA law now allows authorities to take action even when cyberattacks are carried out from outside the country, and they also introduce stronger punishments for serious cyber offences. In addition to criminal laws that punish hackers, other regulations work alongside them to improve cybersecurity protection and ensure organizations follow proper security practices. Complementing criminal law, the Data protection Act 2018 implements **the general data protection law (GDPR)** in the UK context, establishes strict obligations for the company that collects user data & controls it incase, if company fails to protect customer data the UK GDPR will heavily fine the administrative.

Furthermore, **The UK cybersecurity act 2019** tells organization to maintain cybersecurity rule seriously by setting rules and regulation to punish disobedient people. This means the

focus is not only on punishing cybercriminals after attacks happen, but also on preventing attacks by making organizations prepare in advance and strengthen their security.

Addressing Cybercrime, data privacy, critical infrastructure protection, and digital civil rights, handle by framework:

- The ETA treats cybercrime mainly as a **criminal offence handled by law enforcement**. It criminalizes hacking, data damage, and online fraud.
- It also gives **legal recognition to electronic records and digital signatures**, ensuring the validity and security of electronic communication.
- Although the Act mentions privacy and personal data concepts, it does not provide a strong or detailed data protection system. Critical infrastructure protection is not clearly addressed.

Information Technology Bill 2075

- The IT Bill increases penalties for cybercrime and proposes **stronger government authority** in cyber investigations.
- It aims to protect individuals' privacy by regulating the **collection, storage, and processing of personal information**.
- However, concerns exist about excessive government control and possible restrictions on digital freedom.

Computer Misuse Act 1990

- The CMA criminalizes **unauthorized access, system interference, and cyberattacks**.
- It has **strong extraterritorial jurisdiction**, meaning offenders can be prosecuted even if attacks come from outside the UK.
- Penalties are severe, especially for serious attacks on national infrastructure.

Data Protection Act 2018

- This law does not directly punish cybercrime. Instead, it focuses on **protecting personal data**.
- It ensures individuals know how their data is collected, stored, used, and shared by organizations.
- It imposes obligations on companies to maintain strong data protection standards.

General Data Protection Regulation

- GDPR protects data privacy using a **rights-based and accountability-driven approach**.
- It gives individuals strong rights over their data and imposes strict duties on organizations.
- It is globally influential and harmonized across many countries.

Network and Information Systems Regulations 2018

- This framework enhances the **prevention, investigation, and reporting of cyber incidents**, especially in essential sectors like energy and healthcare.
- It improves national cybersecurity readiness and resilience.

Case study – Driven Jurisprudence:

Nepal case:

Mangaluru police arrest 11 Indians in Nepal.

Feb 05, 2026 published by THE HINDU

The Mangaluru city police arrest 11 Indians & The Nepal police have taken custody of some

Chinese nationals who were alleged associates of frauds that were charged for online investment fraud, digital fraud etc by phishing innocent people through social media such as Instagram, WhatsApp etc telling them to invest money on the app & when innocent people transferred their money the scammer will scam the money by transferring it to different banks and then exchange those money into crypto currency & block them.

Analysis on legal principles were applied & discuss.

The law treated it as hacking and frauding. Police used digital evidence and bank records. Courts punished the offenders but weak cross-border rules showed gaps in handling international cybercrime.

UK case:

<u>WannaCry ransomware</u>

In 2017, the WannaCry ransomware attack affected hospitals, businesses, and government systems across the UK and globally. It encrypted data and demanded ransom payments and disrupting critical services like the NHS. The UK response involved rapid incident management, international cooperation, and public guidance. Legal and regulatory frameworks including the Computer Misuse Act 1990 and NIS Regulations highlighted systemic vulnerabilities. The case emphasized the importance of proactive cybersecurity, timely patching, critical infrastructure protection, and corporate accountability in national cyber defense.

Figure 3: WannaCry 2017 Ransomware Attack - UK Legal Response Analysis (AI-generated diagram)

Analysis on legal principles were applied & discuss.

The law punished the offenders for hacking and fraud. Investigations used digital evidence and cooperation with banks. Courts focused on criminals, not on banks and the weak rules in border law has created problem for police to arrest criminal due to weak border rule in different country.

Legal limitations and Strategic Reform:

Enforcement gap in Nepal cybersecurity.

1. Nepal lacks an independent data protection authority
2. Nepal has weak critical infrastructure framework
3. The handling of digital evidence remains underdeveloped, risking the potentially crucial evidence that could influence the outcome of a case.
4. Cyber policing training and technical resources remain insufficient
5. Limited use of word in social according to IT bill 2075

Propose reform strategies.

1. Adapt a risk-based cybersecurity framework of NIST model that guides organization to identify, protect, detect, Respond & recover.

Figure 4: NIST Cybersecurity Framework - Proposed Reform Model for Nepal (AI-generated diagram)

2. Create an official similar to the information commissioner's office to enforce data privacy laws, monitor compliance, impose meaningful penalties for breaches etc.
3. Introduce mandatory Breach reporting for cyber security incident.

4. Define and protect critical infrastructure (NIS2 principles) to identify sector.
5. Developing standard operating operation for cross -border operation and evidence sharing.

TASK 2: AI Threat Detection – Ethical Dilemmas

1. Introduction

AI has become a core part of how organisations defend their networks in cyberworld. Tools built by companies like IBM and Microsoft automatically detect threat detection, flag phishing, identify ransomware, and monitors unusual behaviour or pattern in the system.

We might think due to ai our world has become simplify but the truth is there are difficulty in security while the world is advancing.

The difficulty is that security effectiveness and ethical responsibility don't always point in the same direction. Constant monitoring, behavioural analysis, and algorithmic decision-making all raise genuine concerns about bias, privacy, and accountability. These aren't abstract worries the AI has already produced concrete harms in other technology contexts, and there's no good reason to assume cybersecurity is different. This essay looks at two of those tensions and considers what utilitarian and deontological frameworks have to say about them.

2. Technology in Focus

AI threat detection builds a model of normal behaviour across a network such as login patterns, file access, device use, communication activity and flags deviations from the use of AI. AI Systems learn continuously in everyday, which improves accuracy over time and allows them to catch novel attack patterns that rule-based tools would miss. Response times will be faster, monitoring is more consistent and the workload on human analysts is reduced in future.

The operational case is strong but those benefits depend on persistent, comprehensive surveillance of everything happening on the network which is exactly where the ethical complications begin.

3. Ethical Dilemma 1: Algorithmic Bias and Discrimination

AI systems learn from data and data carries the marks of the world that produced it. when training datasets reflect existing biases, the system learns to reproduce them. Predictive policing tools like PredPol showed this clearly: they ended up flagging minority and lower-

income communities at higher rates not because the risk was objectively greater but because decades of unequal policing had shaped the underlying data.

In cybersecurity, the same dynamic plays out because a biased model might consistently flag users from certain region might, apply tighter restrictions to employees from particular backgrounds or treat behavioural patterns common to specific groups as inherently suspicious. The person affected often has no way of knowing a decision was made about them, let alone challenging it.

Accountability is a separate but related problem. When an algorithm produces a biased outcome, responsibility is spread across developers, deploying organisations, data collectors, and regulators. That diffusion makes the same problems continue far longer than they should.

4. Ethical Dilemma 2: Surveillance vs Privacy

Effective AI threat detection requires watching everything in and outside the system. Login times, browsing activity, file access, communication metadata all of it goes into surveillance. From a security standpoint that dealing with those stuff is important in security point to detect unusal activities but from a privacy standpoint, it represents a significant and often un respectful intrusion into people's working lives.

State-level examples make the stakes visible. China's AI-powered mass surveillance and the UK police's use of facial recognition at protests illustrate how quickly security infrastructure can shift from detecting genuine threats to monitoring lawful behaviour and chilling democratic participation. At the organisational level, the same thing happens more quietly: workplace AI security tools collect extensive personal data under legal frameworks broad enough to authorise almost anything and most employees have no real idea what's being collected or how it's used.

The key question isn't whether any monitoring is acceptable because most people would accept some level of it. It's whether current practice is balanced to actual risk and whether individuals retain any meaningful say. On both counts, the honest answer is usually no.

5. Ethical Evaluation: Utilitarianism

Utilitarianism judges actions by whether they produce more benefit than harm overall. For AI threat detection, the benefits are real such as prevented cyberattacks, protected infrastructure, reduced financial losses. If a system stops a ransomware attack on a hospital's patient records, the outcome is good for a large number of people

but utilitarianism requires the harms to be counted honestly too. Biased systems produce unfair outcomes. people change how they act when they know they're being watched, and that suppression of free thought and expression carries genuine social costs. A proper utilitarian analysis doesn't automatically recommend AI security tools but it demands that benefits and harms be weighed rigorously and continuously, with real willingness to impose limits when the balance tips the wrong way.

6. Ethical Evaluation: Deontological Ethics

Deontological ethics, most closely associated with Kant, holds that some things are wrong regardless of their outcomes specifically, actions that fail to respect the dignity and rights of persons. From this standpoint, the question isn't only whether AI security systems produce good results, but whether the way they treat people is compatible with treating those people as ends rather than means.

Continuous monitoring without meaningful consent treats individuals as data sources. It often operates without any mechanism for people to understand what's being collected about them or to contest decisions made on that basis. Deontological principles require explainability and proportionality — intrusions on privacy should be no greater than what the security objective genuinely requires. Where AI systems don't meet that standard, the ethical response is to constrain them, not to justify the overreach on grounds of effectiveness.

TASK 3: Leadership, Security Culture and Professional Portfolio

Leadership and Organizational Security Culture: A Deep Evaluation

Leader is the important role in every major companies which means they are important in cyber defences also for their high skill in managing the group in the company beyond setting policies, leaders signal priorities, shape risk perception, and influence employee behaviour etc they are more like a team composition person that gives direction to members for the benefit of their organisation. Leader action, communication style, and risk posture create an environment of accountability and vigilance.

Figure 1: The Role of Leadership in Building Organizational Security Culture

Weak Leadership Case Study – Equifax 2017 Breach

The Equifax breach of 2017 remains a primary example of poor cybersecurity leadership in the entire history of the United States, where they suffered a significant data breach holding personal information of nearly half of the US population. In this data breach occurrence between May and July 2017 investigation found that the breach exploited a vulnerability in the Apache Struts web application framework and also found that employees lacked guidance and awareness, reflecting mismanagement and weak leadership.

In conclusion, weak leadership directly contributed to low incident response maturity, ineffective security training and ethical lapses in handling sensitive data.

Strong Leadership Case Study – Microsoft and IBM

Conversely, Microsoft and IBM exemplify the benefits of proactive and ethically-driven leadership these, companies have invested heavily in computer technology and security to build strategic cooperation to establish functional teams and maintaining clear communication during incidents. These kinds of teams operate under structured frameworks that solve threat detection and mitigation under strong leadership.

In conclusion, strong and proactive leadership creates a culture of resilience, vigilance and accountability throughout the entire organization.

Figure 3: Microsoft & IBM — Models of Strong Cybersecurity Leadership

Impact of Leadership on Security Culture

- Prioritizing Threat Mitigation and Incident Response: Leaders cooperate with employees to test services and adopt pen testing to ensure defences.
- Ethical Resilience: By embracing advanced services like SOC as a Service and managed network security, executives ensure robust protection across all layers of the organisation's IT environment.
- Security Awareness Training Effectiveness: Leaders bring training and analysis to employees so that they can be aware of the current needs of knowledge in the security world.

In essence, leadership is not just a managerial function but a cultural force that shapes attitudes toward risk, ethics, and accountability across the organisation.

Integrated Cybersecurity Ethics and Compliance Policy

A robust cybersecurity ethics policy is essential for multinational organisations operating across the globe to ensure that cybersecurity practices are not only compliant with legal requirements but also aligned with ethical standards.

Key benefits of an integrated cybersecurity ethics and compliance policy include:

- Ensuring a clear ethical framework with guidance for all staff
- Implementing effective risk management strategies
- Enabling fast incident response and clear reporting procedures

Proposed Cybersecurity Ethics and Governance Framework

In my opinion, jurisdictions must adopt a unified cybersecurity ethics and governance framework across globe because this will help countries coordinate together as a team for example, having the same law will make it possible to arrest criminals even if they are outside country borders.

Figure 3: Proposed Unified Cybersecurity Ethics & Governance Framework (GDPR, NIST, ISO 27001, ETA 2063)

1. Regulatory Compliance

Nepal: Adhere to the Electronic Transactions Act (ETA) 2063, which governs electronic records, authentication and penalties for cyber offences.

UK and EU: Ensure GDPR compliance, particularly around personal data protection, consent management and cross-border transfers.

International Standards: Align with ISO/IEC 27001, COBIT 2019, and NIST CSF to establish structured risk management, governance and incident response protocols.

2. Ethical Principles

- Least Privilege: Access rights are limited strictly to role requirements, reducing exposure to insider threats.
- Privacy by Design and Data Minimisation: Security and privacy controls are embedded into system design; only essential data is collected and retained.
- Incident Reporting and Transparency: Employees report breaches or near misses without fear of retaliation, ensuring timely and ethical incident management.
- Cross-Border Data Transfer Compliance: Legal mechanisms such as Standard Contractual Clauses (SCCs) or Binding Corporate Rules (BCRs) are used for lawful data transfers.

3. Role-Specific Ethical Guidelines

- SOC Teams: Monitor systems, detect threats, and handle sensitive data ethically, avoiding misuse of privileges.

- CISOs: Lead cybersecurity strategy, enforce compliance, and act as ethical exemplars in decision-making.
- IT Auditors: Maintain independence, objectively evaluate controls, and report compliance and ethical g

TASK 4: Self-Reflection and Professional Ethical Portfolio

1. Self-Reflection on Cybersecurity Skills and Ethical Growth

Cybersecurity is not only a technical discipline but also a profession grounded in responsibility, trust, and ethical judgment. In the modern world, cybersecurity skills and ethical growth are growing rapidly due to a large number of malicious organizations that work in groups to attack private institutions. This has led cybersecurity knowledge to become a critical requirement in today's professional landscape. If an organization fails to protect the data of individual users, it may face significant legal penalties which might potentially could lead to bankruptcy. Having insightful knowledge of cyber skills can secure these organizations and benefit users.

As I reflect on my journey as an aspiring cybersecurity policy advisor, I have discovered that becoming a master in cybersecurity requires more than mastering tools and techniques. It requires a strong ethical foundation and a commitment to continuous improvement. During my academic journey, three key competencies stand out as essential for my growth: digital forensics capability, legal and regulatory compliance awareness, ethical incident response combined with data privacy understanding etc.

1.1 Digital Forensics

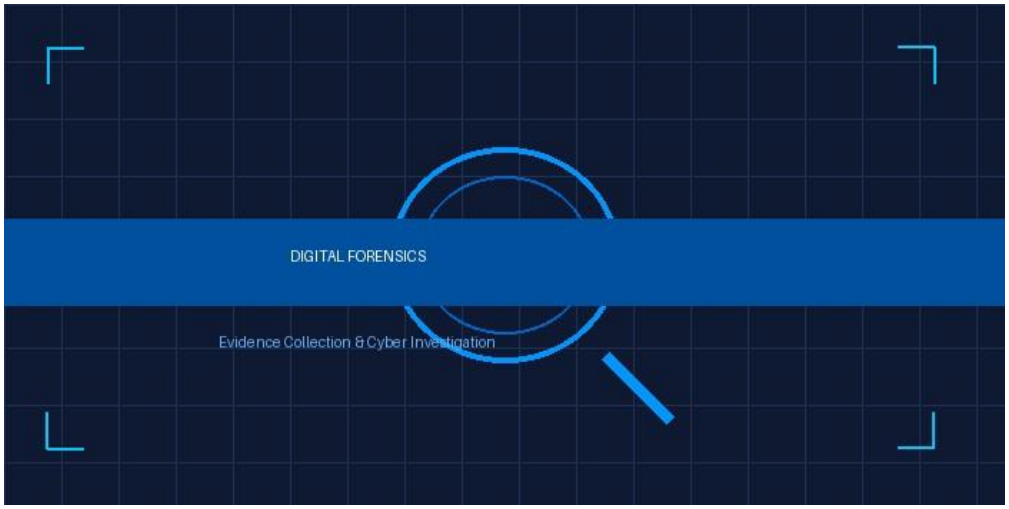

Figure 1: Digital Forensics – Evidence Collection and Cyber Investigation

The first critical is digital forensics because in today's world the threat of cyber attacks are everywhere detecting a cyberattack is only the beginning because understanding how it occurred and preserving evidence correctly is equally important. Digital forensics ensures that investigations are accurate, legally defensible, and ethically conducted. Through academic learning, I have been exposed to analyzing logs, examining packet captures and

identifying indicators of compromise. These activities help understand how attackers leave traces and how investigators reconstruct events.

After building my academic foundation, I plan to construct a controlled home lab environment where I can simulate attacks and practice forensic investigation using open-source tools. Participating in Capture the Flag (CTF) competitions focused on forensic challenges will further strengthen my analytical thinking under time pressure.

1.2 Legal and Regulatory Compliance

Figure 2: Legal Compliance Frameworks – GDPR and Nepal's Electronic Transactions Act 2063

Secondly, understand the legal frameworks governing digital systems and personal data of Nepal, the Electronic Transactions Act (ETA) 2063 serves as the foundation for regulating electronic transactions and cybercrime if we managed to learn the ETA 2063 then alongside this, the General Data Protection Regulation (GDPR) of the UK and EU should be the target to learn because this law covers most of the cyber law in the world which helps securities to adapt other country law as fast as possible.

Understanding these frameworks allows cybersecurity professionals to ensure that their practices are legally compliant. This is particularly important for incident response, where actions taken during an investigation must adhere to both national and international legal obligations, including consent, data minimization, and transparency requirements.

1.3 Ethical Incident Response and Data Privacy

The third competency is ethical incident response combined with data privacy awareness. Cyber incidents often demand quick decisions under significant pressure. During such situations, cybersecurity professionals must balance organizational protection with respect

for individual rights. Platforms such as TryHackMe provide practical, hands-on training for developing these skills in realworld simulated environments.

2. Building a Professional Ethical Portfolio

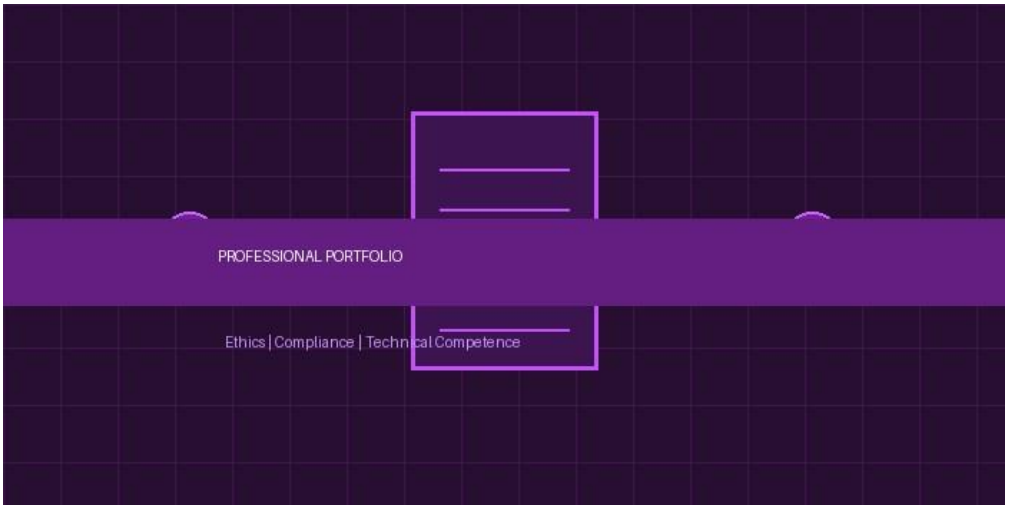

Figure 3: Building a Professional Ethical Portfolio in Cybersecurity

When graduating from an academic programme, many students are uncertain about the next steps as junior cybersecurity professionals. In my opinion, a cybersecurity professional can construct a credible portfolio by aligning technical competence with ethical responsibility and legal awareness.

2.1 Ethical Decision-Making Under Pressure

Ethical decision-making can be showcased through ethical hacking projects, forensic Capture The Flag (CTF) participation, and red-team/blue-team simulations. Documenting proper authorization, responsible disclosure, and recommended remediation demonstrates maturity and accountability beyond technical execution. Certifications such as the Certified Ethical Hacker (CEH) further validate ethical hacking proficiency. Leading workshops or mentoring peers in controlled environments also strengthens leadership skills and ethical awareness.

2.2 Legal and Regulatory Compliance Documentation

Portfolios can include drafted data protection policies aligned with Nepal's Electronic Transactions Act 2063 and documentation showing adherence to international frameworks such as the GDPR — including consent, transparency, and data minimization practices. Privacy impact assessments, compliance checklists, and data flow diagrams provide evidence of understanding and implementing national and international legal obligations.

2.3 Technical Depth and Demonstration

Technical depth must be clearly highlighted. GitHub repositories containing secure authentication systems, log analysis scripts, and penetration testing reports demonstrate analytical skills and structured methodology. Digital forensic case studies involving packet capture and evidence preservation illustrate investigative competence and attention to detail.

2.4 Ethical Branding and Reputation Management

Ethical branding and reputation management are critical for long-term career growth. Maintaining a professional online presence, responsibly sharing knowledgeand disclosing vulnerabilities transparently build trust and credibility. In cybersecurity, reputation acts as a professional asset because employers value individuals who combine technical expertise with integrity.

Overall, a well-structured portfolio that integrates ethics, compliance and technical competence builds long-term trust and supports sustainable career growth in cybersecurity.

3. Conclusion:

In this assignment there are 4 different task submission which having different lesson to learn such as task 1 covers Comparative Cybersecurity Legal Frameworks where we learn that if we want to be in the world of cyber security then we must learn the laws and rules regulation of cyber world where we talk about UK, Nepal and EU ethical law. These Nepal law had some major flaws in ETA 2063 then the Nepal wanted to establish new law to solve & fix those error that occurred in ETA 2063 but the law was unable to be pass due to controversy and Genz protest. We also learned about the advance law of cyber law that where applied in Eu and UK the task 2 covers the ethical dilemmas where we talk about the benefit & disadvantage of AI covering it with Utilitarianism and Deontological Ethics the task 3 covers the importance of leadership & its role in organization and task 4 covers the ethical portfolio and Self-Reflection on Cybersecurity Skills and Ethical Growth.

In my opinion, my suggestion for my country Nepal to be developed is if we established the globe cyber law then we could potential mitimize the rising trend of hacking by allowing police officer to investigate outside of the country also without borderline disturbance after this, now lets talk about the ai where Ai has showed biasness making people be in a situation of ethically not allowed position so to minimize this company should follow the new rule of cybersecurity and take accountability of privacy attack along with law breaking rule. That it in my opinion which should be improved for Nepal cyber security to be focused.

References

- CEH Certification by EC-Council: https://www.eccouncil.org/train-certify/certified-ethical-hacker-ceh
- CEH – AI-Powered Ethical Hacking: https://ictframe.com/ai-powered-ethical-hacking-certification/
- Electronic Transaction Act 2063 – Explanation: https://mokshalegalgroup.com/electronictransaction-act/
- ETA Cybercrime Penalties: https://bimstudies.com/docs/it-ethics-and-cybersecurity/cyber-law-incontext-of-nepal/electronic-transaction-act/
- Official EU GDPR Resources: https://gdpr.eu/what-is-gdpr/

TryHackMe – Cybersecurity Training: https://tryhackme.com

Floridi, L. et al. (2018) 'AI4People — An Ethical Framework for a Good AI Society'. Minds and Machines, 28(4), pp. 689–707. Available at: https://link.springer.com/article/10.1007/s11023018-9482-5 (Accessed: 10 February 2026)

Mittelstadt, B.D. et al. (2016) 'The Ethics of Algorithms: Mapping the Debate'. Big Data & Society, 3(2), pp. 1–21. Available at: https://journals.sagepub.com/doi/10.1177/2053951716679679 (Accessed: 10 February 2026)

O'Neil, C. (2016) Weapons of Math Destruction: How Big Data Increases Inequality and Threatens Democracy. New York: Crown Publishers. Available at: https://www.penguinrandomhouse.com/books/241363/weapons-of-math-destruction-bycathy-oneil/ (Accessed: 11 February 2026)

Rachels, J. and Rachels, S. (2019) The Elements of Moral Philosophy. 9th edn. New York: McGrawHill. Available at: https://www.mheducation.com/highered/product/elements-moralphilosophy-rachels-rachels/M9781259914256.html (Accessed: 11 February 2026)

Russell, S. and Norvig, P. (2020) Artificial Intelligence: A Modern Approach. 4th edn. Hoboken: Pearson. Available at: https://aima.cs.berkeley.edu/ (Accessed: 12 February 2026)

Taddeo, M. and Floridi, L. (2018) 'The Debate on the Moral Responsibilities of Online Service Providers'. Science and Engineering Ethics, 22(6), pp. 1575–1603. Available at: https://link.springer.com/article/10.1007/s11948-014-9616-6 (Accessed: 12 February 2026)

Whittlestone, J. et al. (2019) 'The Role and Limits of Principles in AI Ethics'. Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society, pp. 195–200. Available at: https://dl.acm.org/doi/10.1145/3306618.3314289 (Accessed: 13 February 2026)

Zarsky, T.Z. (2016) 'The Trouble with Algorithmic Decisions'. Science, Technology & Human Values, 41(1), pp. 118–132. Available at: https://journals.sagepub.com/doi/10.1177/0162243915605575

Equifax (2017). 2017 Equifax Data Breach. Wikipedia. Available at: https://en.wikipedia.org/wiki/2017_Equifax_data_breach

Seven Pillars Institute. Equifax Cybersecurity Governance Weaknesses Case Study. Available at: https://www.sevenpillarsinstitute.org/case-study-equifax-data-breach/

Microsoft Security Blog (2025). Building a Lasting Security Culture at Microsoft. Available at: https://www.microsoft.com/en-us/security/blog/2025/10/13/building-a-lasting-security-culture-atmicrosoft/

Microsoft Security Blog (2025). Build a Stronger Security Strategy with Proactive and Reactive Incident Response. Available at: https://www.microsoft.com/en-us/security/blog/2025/02/10/builda-stronger-security-strategy-with-proactive-and-reactive-incident-response-cyberattack-series/

Sprintzeal (2024). IBM Cybersecurity Strategies and Defences Overview. Available at: https://www.sprintzeal.com/blog/ibm-data-breach

Palo Alto Networks. Cybersecurity Compliance and Regulatory Frameworks Explained. Available at: https://www.paloaltonetworks.com/cyberpedia/cybersecurity-compliance-and-regulations

GDPR Local. Comparing Information Security Frameworks and Data Protection Frameworks. Available at: https://gdprlocal.com/comparing-information-security-frameworks-and-dataprotection-frameworks/

2063 (ETA) including cybercrime provisions:
https://niice.org.np/archives/4061

cyber offenses and penalties under, ETA 2063:
https://corporatebizlegal.com/electronic-transaction-act-nepal-2063/

digital signatures and electronic records:
https://notarynepal.com/blog/electronic-transaction-act-nepal

Report on case of Mangaluru police:
https://www.ndtv.com/karnatakanews/mangaluru-police-busts-multi-crore-global-cyber-fraud-11arrested-10956639

Alternative article on the same Nepal-linked cyber fraud operation:
https://indianexpress.com/article/cities/bangalore/mangaluru-policebust-nepal-investment-fraud-syndicate-11-arrested-10516973/

UK Data Protection Act 2018 overview:

https://www.legislation.gov.uk/ukpga/2018/12/contents

(UK GDPR + Data Protection Act): https://www.gov.uk/data-protection

UK Computer Misuse Act 1990:
https://en.wikipedia.org/wiki/Computer_Misuse_Act_1990

UK cybersecurity laws covering CMA, DPA, and Network and Information Systems Regulations: https://www.minstermicro.co.uk/information-security-laws-in-the-uk/

Article on what GDPR is and how it protects data privacy:
https://www.wired.com/story/what-is-gdpr-uk-eu-legislation-compliancesummary-fines-2018